

```
here <- $(x,0)$[]
there <- $(x,10)$[]
gridx <- $10$[]
gridy <- $10$[]
resolution <- $16$[]
```

```
BLOCK unreachable0 ()
[]
{
  STOP
}
```

```
BLOCK entry ()
[]
{
  gridx <- ${0} - 1$[gridx];
  gridy <- ${0} - 1$[gridy];
  step <- $(( ${0}.x - {1}.x ) / {2}, ( ${0}.y - {1}.y ) / {2}) :: point$[there, here, resolution];
  loc <- ${0}$[here];
  max_angle <- $NULL :: float$[];
  i <- $1$[];
  GOTO inter0 ()
}
```

```
BLOCK inter0 ()
[]
{
  hhere <- $SELECT SUM(( ${0}! / ((c.x!) * (( ${0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ( ${0} - c.x)) *
    ( ${1}! / ((c.y!) * (( ${1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ( ${1} - c.y))) AS h
    FROM controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)$[gridx, gridy, here];
  GOTO inter1 ()
}
```

```
BLOCK inter1 ()
[]
{
  GOTO inter2 ()
}
```

```
BLOCK inter2 ()
[]
{
  GOTO inter3 ()
}
```

```
BLOCK inter3 ()
[]
{
  GOTO inter4 ()
}
```

```
BLOCK inter4 ()
[]
{
  GOTO inter5 ()
}
```

```
BLOCK inter5 ()
[]
{
  GOTO inter6 ()
}
```

```
BLOCK inter6 ()
[]
{
  JUMP loop0_head ()
}
```

```
BLOCK loop0_head ()
[]
{
  condition0 <- ${0} > {1}$[i, resolution];
  GOTO inter7 ()
}
```

```
BLOCK inter7 ()
[]
{
  IF condition0
  THEN GOTO truthy0 ()
  ELSE GOTO falsey0 ()
}
```

```
BLOCK truthy0 ()
[]
{
  EMIT ${0} = {1}$[angle, max_angle];
  GOTO loop0_exit ()
}
```

```
BLOCK falsey0 ()
[]
{
  loc <- $({0}.x + {1}.x, {0}.y + {1}.y) :: point$[loc, step];
  GOTO inter8 ()
}
```

```
BLOCK loop0_exit ()
[]
{
  GOTO inter13 ()
}
```

```
BLOCK inter8 ()
[]
{
  hloc <- $SELECT SUM(( ${0}! / ((c.x!) * (( ${0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ( ${0} - c.x)) *
    ( ${1}! / ((c.y!) * (( ${1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ( ${1} - c.y))) AS h
    FROM controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)$[gridx, gridy, loc];
  GOTO inter9 ()
}
```

```
BLOCK inter13 ()
[]
{
  STOP
}
```

```
BLOCK inter9 ()
[]
{
  angle <- $degrees(atan(( ${0} - {1}) / sqrt(( {2}.x - {1}.x)** 2 + ( {2}.y - {1}.y)** 2)))$[hloc, hhere, loc];
  GOTO inter10 ()
}
```

```
BLOCK inter10 ()
[]
{
  condition1 <- ${0} IS NULL OR {1} > {2}$[max_angle, angle, max_angle];
  GOTO inter11 ()
}
```

```
BLOCK inter11 ()
[]
{
  IF condition1
  THEN GOTO truthy1 ()
  ELSE GOTO falsey1 ()
}
```

```
BLOCK truthy1 ()
[]
{
  max_angle <- ${0}$[angle];
  GOTO inter12 ()
}
```

```
BLOCK inter12 ()
[]
{
  GOTO merge1 ()
}
```

```
BLOCK falsey1 ()
[]
{
  GOTO merge1 ()
}
```

```
BLOCK merge1 ()
[]
{
  GOTO merge0 ()
}
```

```
BLOCK merge0 ()
[]
{
  JUMP loop0_head ()
}
```